

Determination of Critical Power Using a 3-min All-out Cycling Test

ANNI VANHATALO¹, JONATHAN H. DOUST², and MARK BURNLEY¹

¹Department of Sport and Exercise Science, University of Wales, Aberystwyth, UNITED KINGDOM; and ²Chelsea School Research Centre, University of Brighton, Eastbourne, UNITED KINGDOM

ABSTRACT

VANHATALO, A., J. H. DOUST, and M. BURNLEY. Determination of Critical Power Using a 3-min All-out Cycling Test. *Med. Sci. Sports Exerc.*, Vol. 39, No. 3, pp. 548–555, 2007. **Purpose:** We tested the hypothesis that the power output attained at the end of a 3-min all-out cycling test would be equivalent to critical power. **Methods:** Ten habitually active subjects performed a ramp test, two 3-min all-out tests against a fixed resistance to establish the end-test power (EP) and the work done above the EP (WEP), and five constant-work rate tests to establish the critical power (CP) and the curvature constant parameter (W') using the work-time and 1/time models. **Results:** The power output in the 3-min trial declined to a steady level within 135 s. The EP was 287 ± 55 W, which was not significantly different from, and highly correlated with, CP (287 ± 56 W; $P = 0.37$, $r = 0.99$). The standard error for the estimation of CP using EP was approximately 6 W, and in 8 of 10 cases, EP agreed with CP to within 5 W. Similarly, the WEP derived from the 3-min test (15.0 ± 4.7 kJ) was not significantly different from, and correlated with, W' (16.0 ± 3.8 kJ; $P = 0.35$; $r = 0.84$). **Conclusions:** During a 3-min all-out cycling test, power output declined to a stable value in approximately the last 45 s, and this power output was not significantly different from the independently measured critical power. **Key Words:** EXERCISE INTENSITY DOMAINS, EXERCISE TESTING, ANAEROBIC CAPACITY, POWER-DURATION RELATIONSHIP, PEAK $\dot{V}O_2$

On the basis of the characteristic blood lactate and oxygen uptake ($\dot{V}O_2$) responses during constant-work rate exercise, three exercise intensity domains have been defined: moderate, heavy, and severe (12,22,27). The boundary between the moderate and heavy domains can be established during incremental exercise testing (28) using the lactate threshold or its gas exchange equivalent (gas exchange threshold, or GET). This threshold is important because it demarcates work rates for which a $\dot{V}O_2$ slow component is present (heavy and severe exercise) from those in which it is not (12,27). The boundary between heavy and severe exercise signifies the highest work rate at which a steady state can be attained; work rates above this boundary result in $\dot{V}O_2$ and blood [lactate] rising as a function of time until the $\dot{V}O_{2\text{peak}}$ is attained, with exhaustion occurring thereafter (12,22,23,27). Identification of the heavy-severe boundary presently requires repeated and often exhaustive exercise testing performed over

several days, yielding the maximal steady-state work rate (if blood [lactate] and/or gas exchange indices are used (3,17,18,26)) or the critical power (CP; if a series of work rates are performed to exhaustion (4,6,16,20,23,25)). Consequently, the heavy-severe domain boundary is not routinely established during exercise testing for research or diagnostic purposes. A test protocol that provides an estimate of the heavy-severe domain boundary in a single laboratory visit would, therefore, be a useful addition to an exercise testing battery.

One method of estimating the heavy-severe domain boundary is to establish the CP (12,15,19,20), which requires a subject to exercise to exhaustion at several constant work rates on separate days. The relationship between power output and time to exhaustion is hyperbolic and is defined by two parameters: CP, which represents the highest sustainable work rate; and the curvature constant (W'), which is the maximum amount of work that can be performed above CP (10,11,22) and is often referred to as anaerobic work capacity (21). Linear formulations of this relationship can be obtained by plotting total work done during the series of severe exercise bouts against time (19) or by plotting power output against the inverse of time (22). The linear 1/time model is given by:

$$P = \frac{W'}{t} + \text{CP} \quad [1]$$

where W' is represented by the slope, and CP is represented by the y-intercept (10,22). During any exercise bout performed above CP, the fixed capacity for work denoted by W' is gradually expended and cannot be replenished

Address for correspondence: Mark Burnley, Department of Sport and Exercise Science, Carwyn James Building, University of Wales, Aberystwyth, Aberystwyth, Ceredigion, SY23 3FD, UK; E-mail: mhb@aber.ac.uk.

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until exercise is terminated or work rate falls below CP (6,21). Equation 1 implies that at the time point where the W' becomes wholly depleted, the highest achievable power output is CP (if $W' = 0$, then $P = CP$). Therefore, it should be possible to use the entire W' in a sufficiently long all-out exercise bout, in which power output would decrease progressively until CP was attained (5). However, this hypothesis has not been tested by comparing the outcome of prolonged all-out exercise against the conventional method of CP determination.

We have previously demonstrated (5) that a 3-min all-out cycling test results in a highly reproducible power profile that levels out during the final 30–60 s at a power output approximately halfway between the lactate threshold (estimated by gas exchange indices) and the peak work rate attained in a ramp test—that is, close to the power output at which the heavy–severe domain boundary would be expected to occur (22,23). As stated previously (5), the 3-min test duration was chosen to provide a protocol that was long enough to yield a stable power output at the end of the test, but not so long that subjects would fail to complete the test. Although the design of a 3-min all-out test was based on the assumptions of the power–duration relationship described above, we did not directly compare the 3-min end-test power (EP) with CP. Moreover, the parameter suggested to represent the anaerobic work capacity (the work done above EP, or WEP) was not validated against an established W' estimate. Recent work has suggested that W' can be estimated from all-out exercise (9), but it is clear that 90 s was an insufficient test duration to yield an EP equal to CP (see Figure 2 in Dekker et al. (9)). This, in addition to the findings of our previous work (5), led us to suppose that the parameters of power–duration relationship could be defined using a 3-min all-out cycling test.

The purpose of the present study was, therefore, to compare the parameters of the power–duration relationship derived from a 3-min all-out cycling test with those derived from a series of five exhaustive exercise bouts for the conventional method of CP determination. Specifically, we aimed to test the hypotheses that 1) the EP in a 3-min all-out cycling test is equivalent with CP, and 2) the WEP in the same test is equivalent to W' .

METHODS

Subjects. Ten subjects (mean $\pm$ SD: age 33 ± 9 yr, body mass 74.1 ± 11.0 kg, height 1.79 ± 0.09 m) gave written informed consent to participate in the study, which was approved by the ethics committee of the University of Wales, Aberystwyth, United Kingdom. All subjects were accustomed to high-intensity exercise and included competitive road cyclists ($N = 6$), club-level distance runners ($N = 2$), and those in general fitness training ($N = 2$). Five subjects had taken part in the previous study, which had been conducted in the same laboratory, using

the same 3-min all-out test protocol as in the current study. Before testing, subjects were informed of the protocol and the possible risks and benefits associated with the project. Subjects were instructed to be rested (no heavy training on the previous day), adequately hydrated, and to have consumed no alcohol for 24 h and no food or caffeine for 3 h before each testing session.

Experimental design. The protocol consisted of eight visits to the laboratory. Subjects had a minimum of 24 h of rest between tests, and all testing was completed within 3 wk. First, the subjects performed an incremental ramp protocol of $30 \text{ W} \cdot \text{min}^{-1}$ for assessment of $\dot{V}O_{2\text{peak}}$ and GET. During the second visit, subjects performed a 3-min all-out familiarization trial, which was not included in the subsequent data analysis. On the following visits, subjects performed a 3-min all-out test to determine EP and WEP, and five predicting trials at constant work rates to exhaustion to determine CP and W' , in a random order. Before each trial, subjects were given a 5-min warm-up at 100 W, followed by a 5-min rest.

Determination of peak oxygen uptake and GET.

All exercise testing was conducted using an electronically braked cycle ergometer (Lode Excalibur Sport, Groningen, The Netherlands). The ergometer seat and handlebars were adjusted for comfort, with the cyclists' own pedals fitted if required, and with the same settings replicated for subsequent tests. The ramp protocol consisted of 3 min of unloaded baseline pedaling, followed by a ramp increase in power output of $30 \text{ W} \cdot \text{min}^{-1}$ until volitional exhaustion. Subjects were instructed to maintain their preferred cadence (80 rpm, $N = 2$; 90 rpm, $N = 8$) for as long as possible. The test was terminated when the pedal rate fell to more than 10 rpm below the chosen cadence for more than 10 s, despite strong verbal encouragement. $\dot{V}O_{2\text{peak}}$ was determined as the highest average $\dot{V}O_2$ during a 30-s period. Data were reduced to 10-s averages for the estimation of GET using the V-slope method (2).

Three-min all-out tests. Subjects first performed a warm-up at 100 W, followed by 5 min of rest. The test started with 3 min of unloaded baseline pedaling at each subject's preferred cadence, followed by an all-out 3-min effort. Subjects were asked to increase their cadence to approximately 110 rpm during the last 5 s of the baseline period. The resistance on the pedals during the 3-min effort was set using the linear mode of the ergometer so that the subjects would attain the power output halfway between $\dot{V}O_{2\text{peak}}$ and the GET (i.e., $\text{GET} + 50\% \Delta$, with Δ being the magnitude of the interval between the GET and $\dot{V}O_{2\text{peak}}$) on reaching their preferred cadence (linear factor = power/cadence squared). Strong verbal encouragement was provided throughout the test, although the subjects were not informed of the elapsed time, to prevent pacing. To ensure an all-out effort, subjects were instructed to maintain their cadence as high as possible at all times throughout the test. Peak $\dot{V}O_2$ was calculated as the highest 30-s average achieved during the test, and blood [lactate]

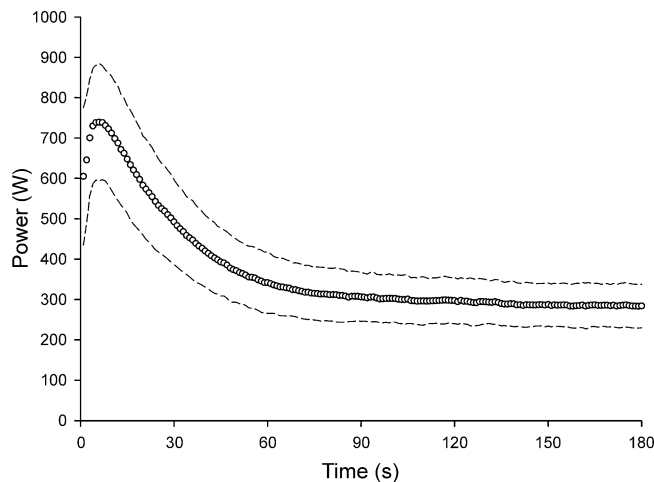


FIGURE 1—The group mean power profile of the 3-min all-out test. Dashed lines indicate the standard deviation. Note the leveling out of the power output during approximately the last 45 s of the test at a power output well below the end ramp power (407 W) and well above the power output associated with the GET (182 W).

was sampled at rest before the test and immediately after its completion (see below).

CP protocol. CP and W' were estimated from five predicting trials. The work rates for the first four trials were equivalent to 70 and 80% Δ and 100 and 105% $\dot{V}O_{2peak}$, and the work rate for the final trial (either 60% Δ or 110% $\dot{V}O_{2peak}$) was selected to obtain a range of times to exhaustion between 2 and 15 min (15). Each trial was preceded by a 5-min warm-up at 100 W and a 5-min rest period, followed by 3 min of unloaded baseline pedaling. Subjects were instructed to maintain their preferred cadence for as long as possible. A test was terminated when cadence fell to more than 10 rpm below preferred cadence for more than 5 s. Strong verbal encouragement was provided throughout the test, and time to exhaustion was recorded to the nearest second. Blood [lactate] was sampled at rest before each trial and immediately after exhaustion. Subjects were not informed

of the work rates or their performance on any of the tests until the entire protocol had been completed. Linear regression was used to provide two sets of CP and W' estimates from the results of these trials, using the work-time ($W = CPt + W'$) and the 1/time ($P = W'(1/t) + CP$) models.

Equipment. For measurement of pulmonary gas exchange, subjects wore a nose clip and breathed through a low-dead space (90 mL), low-resistance (0.75 mm Hg·L⁻¹·s⁻¹ at 15 L·s⁻¹) mouthpiece and impeller turbine assembly (Jaeger Triple V). The inspired and expired gas volume and gas concentration signals were continuously sampled at 100 Hz, the latter using paramagnetic (O₂) and infrared (CO₂) analyzers (Jaeger Oxycon Pro, Hoechberg, Germany) via a capillary line connected to the mouthpiece. These analyzers were calibrated before each test with gases of known concentration, and the turbine volume transducer was calibrated using a 3-L syringe (Hans Rudolph, MO). The volume and concentration signals were time aligned by accounting for the delay in capillary gas transit and analyzer rise time relative to the volume signal. Oxygen uptake, carbon dioxide output, and minute ventilation were calculated using standard formulae (1) and were displayed breath-by-breath. Capillary blood samples were analyzed for whole-blood [lactate] using an automated blood analyzer (YSI 2300 Stat Plus, Yellow Springs, OH).

Data analysis. The all-out test EP was calculated as the average power output for the final 30 s of the test, and the WEP was calculated as the power-time integral above end power. The comparisons between three sets of parameter estimates (work-time model, 1/time model, and the all-out test) were analyzed using a one-way ANOVA with repeated measures, with specific differences identified using 95% paired-samples confidence intervals. Correlation coefficients and bias $\pm$ 95% limits of agreement were used to assess the relationships between the 3-min EP and CP, and WEP and W' . Paired-samples *t*-tests were used to compare the peak $\dot{V}O_2$ in the 3-min test and $\dot{V}O_{2peak}$ in the ramp test, as well as the predicted and

TABLE 1. Comparison of peak $\dot{V}O_2$ response and parameters of the power-duration relationship derived from the 3-min all-out test with conventional measurements.

Subject	$\dot{V}O_{2peak}$		Critical Power Estimates			W' Estimates		
	Ramp $\dot{V}O_{2peak}$ (L·min ⁻¹)	3-min Test $\dot{V}O_{2peak}$ (L·min ⁻¹)	3-min Test EP (W)	Work-Time Model CP (W)	1/Time Model CP (W)	3-min Test Model WEP (kJ)	Work-Time Model W' (kJ)	1/Time Model W' (kJ)
1	2.69	2.73	173	177	176	9.3	9.7	9.9
2	4.53	4.23	286	287	296	16.0	18.9	16.7
3	4.79	4.83	325	337	343	25.3	22.1	20.5
4	4.89	4.60	351	347	347	12.2	11.2	11.3
5	3.70	3.63	252	247	245	18.0	16.7	17.1
6	3.80	3.57	262	251	254	12.2	18.1	16.9
7	4.20	4.21	316	317	320	16.2	17.4	16.5
8	4.71	4.30	334	334	332	11.5	14.9	15.6
9	4.50	4.51	330	333	334	18.1	18.4	18.2
10	4.03	3.90	240	242	247	11.3	12.7	11.0
Mean	4.18	4.05*	287	287	289	15.0	16.0	15.4
SD	0.66	0.61	55	56	57	4.7	3.8	3.5

EP, end-test power output; WEP, work done above end-test power output; CP, critical power.

* Significantly different from ramp $\dot{V}O_{2peak}$ ($P < 0.05$). No significant differences between the parameters of the power-duration relationship derived from the 3-min all-out test and those of the CP models were observed.

actual times to exhaustion in the predicting trials. Statistical significance was accepted at $P < 0.05$ level, with data presented as means $\pm$ SD.

RESULTS

The mean ramp-test $\dot{V}O_{2\text{peak}}$ was $4.18 \pm 0.66 \text{ L}\cdot\text{min}^{-1}$, maximum ramp-test power was $407 \pm 67 \text{ W}$, and the GET

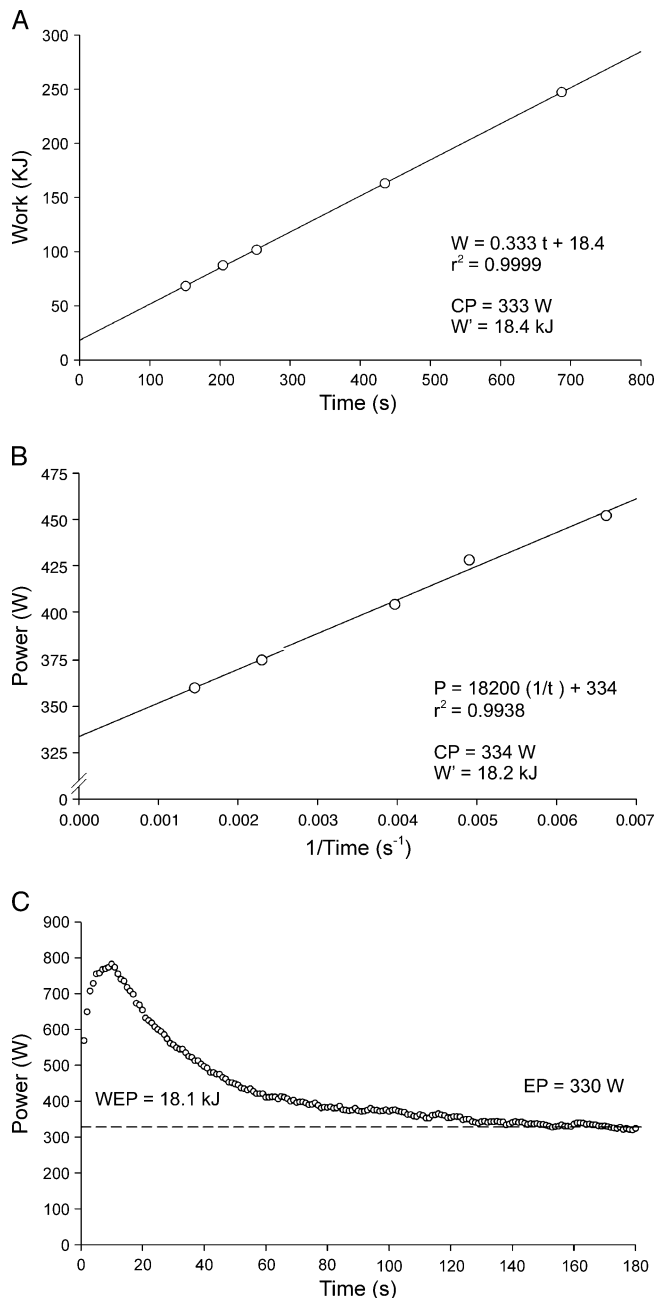


FIGURE 2—The derivation of the critical power (CP) and the curvature constant (W') estimates from the linear work–time (A) and the 1/time (B) CP models, and a 3-min all-out test power profile of a representative subject (C). Note the similarity of CP estimates using all three methods. Also note that, in this subject, WEP is similar to W' .

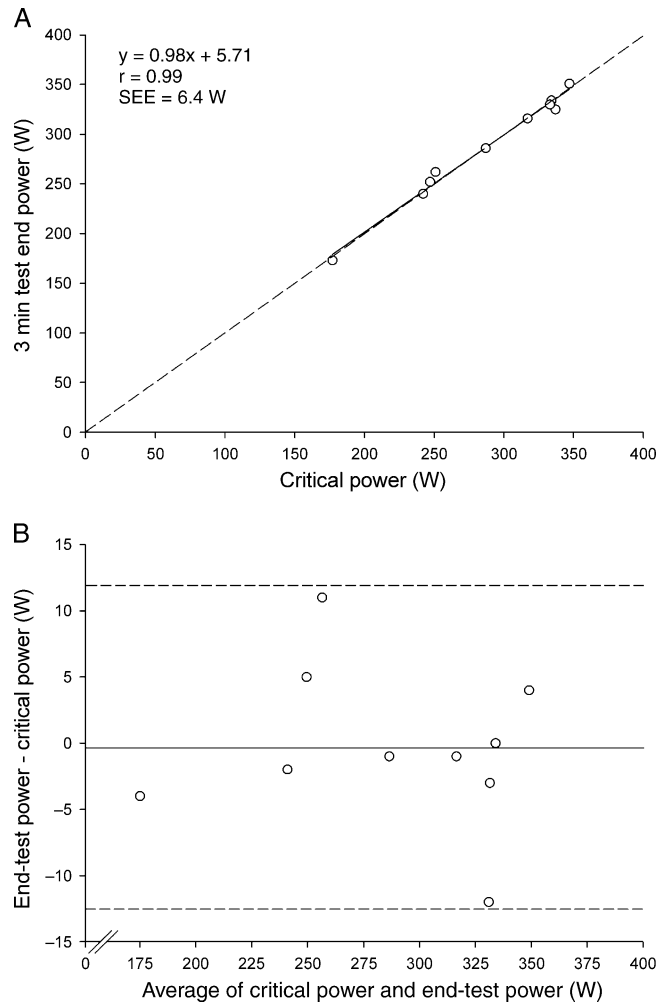


FIGURE 3—Bland–Altman plots of the relationship (A) and limits of agreement (B) between critical power and end-test power. In panel B, the solid horizontal line represents the mean difference between end-test power and critical power, and the dashed lines represent the 95% limits of agreement.

was $2.60 \pm 49 \text{ L}\cdot\text{min}^{-1}$ ($182 \pm 39 \text{ W}$). The peak $\dot{V}O_2$ of $4.05 \pm 0.61 \text{ L}\cdot\text{min}^{-1}$ measured in the 3-min all-out test was significantly lower than the ramp-test $\dot{V}O_{2\text{peak}}$ ($\sim 97\% \dot{V}O_{2\text{peak}}$; $t = 2.60$, $P = 0.029$).

The peak power output in the 3-min test, typically attained within 5–10 s of the start of the test, was $758 \pm 134 \text{ W}$ ($\sim 188\%$ max ramp power; $\sim 141 \text{ rpm}$). The power output during the final 30 s of the test (EP) was $287 \pm 55 \text{ W}$ ($\sim 87 \text{ rpm}$), which represented approximately 70% maximum ramp-test power, approximately 160% GET, and approximately 46% Δ . The WEP was $15.0 \pm 4.7 \text{ kJ}$, or $202 \pm 50 \text{ J}\cdot\text{kg}^{-1}$. The mean power profile of the 3-min test is shown in Figure 1. When the power output data were reduced to 15-s averages and compared, a significant decrease was detected from one time bin to the next ($F_{11,9} = 128.9$, $P < 0.001$), with the exception of the final 45 s, where the changes in power output were -2 W (95% confidence limits $-5.1, 1.1 \text{ W}$) and 0 W (95% CL $-4.8, 4.0$) for the last two comparisons. Thus, power output had stabilized in the last 30 s of the test,

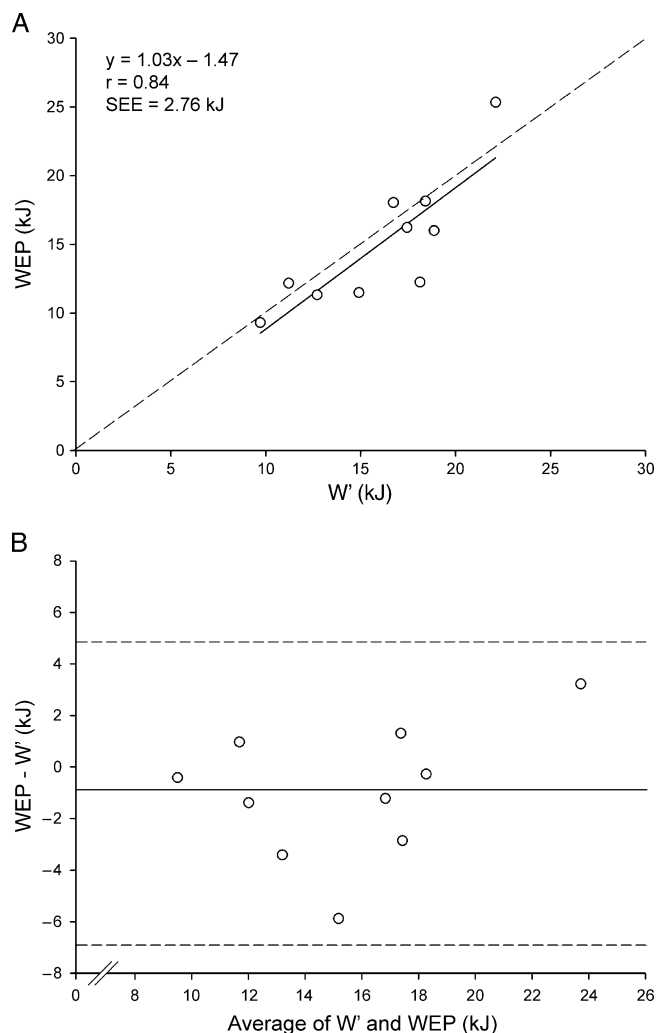


FIGURE 4—Bland-Altman plots of the relationship (A) and limits of agreement (B) between the curvature constant parameter (W') and the work done above the end power (WEP). In panel B, the solid horizontal line represents the mean difference between WEP and W' , and the dashed lines represent the 95% limits of agreement.

justifying the 30-s average being used as EP. Blood [lactate] at the end of exercise was 10.2 ± 2.2 mM, and the total work done in 3 min was 66.2 ± 12.6 kJ.

CP was 287 ± 56 W and W' was 16.0 ± 3.8 kJ when the results of the five predicting trials were modeled using the work-time model, and 289 ± 57 W and 15.4 ± 3.5 kJ when using the 1/time model (Table 1). The CP estimates derived using the two models correlated well and were not significantly different from the all-out test EP ($F_{2,9} = 0.97$, $P = 0.37$) or the two W' estimates from the WEP ($F_{2,9} = 1.03$, $P = 0.35$). Figure 2 demonstrates the derivation of the parameter estimates using the work-time and 1/time models and the power profile of a 3-min all-out test of one of the subjects. Although there were no differences in the parameter estimates from the different CP models, and the correlation coefficients were very high for both models (work-time model range, $r = 0.966$ – 1.00 ; 1/time model range, $r = 0.955$ – 0.999), the work-time model generally fit

the data better and was, therefore, used for further analysis. The parameter estimates correlated well between the two models ($r = 1.00$ for CP; $r = 0.99$ for W'), indicating that there was no systematic error in the predicting data. Figure 3 illustrates the relationships and bias $\pm 95\%$ limits of agreement between EP and CP. The correlation coefficient for EP and CP was $r = 0.99$, and the standard error of the estimate was approximately 6 W (Fig. 3). The same assessment of WEP and W' , presented in Figure 4, produced a correlation coefficient of $r = 0.84$ and a standard error of the estimate of approximately 2.8 kJ.

Figure 5 illustrates the power-duration relationship plotted using the parameters derived from the 3-min all-out test (EP and WEP), along with the actual tolerable durations of all five predicting trials used to determine CP in two different subjects. In panel A, the predicted

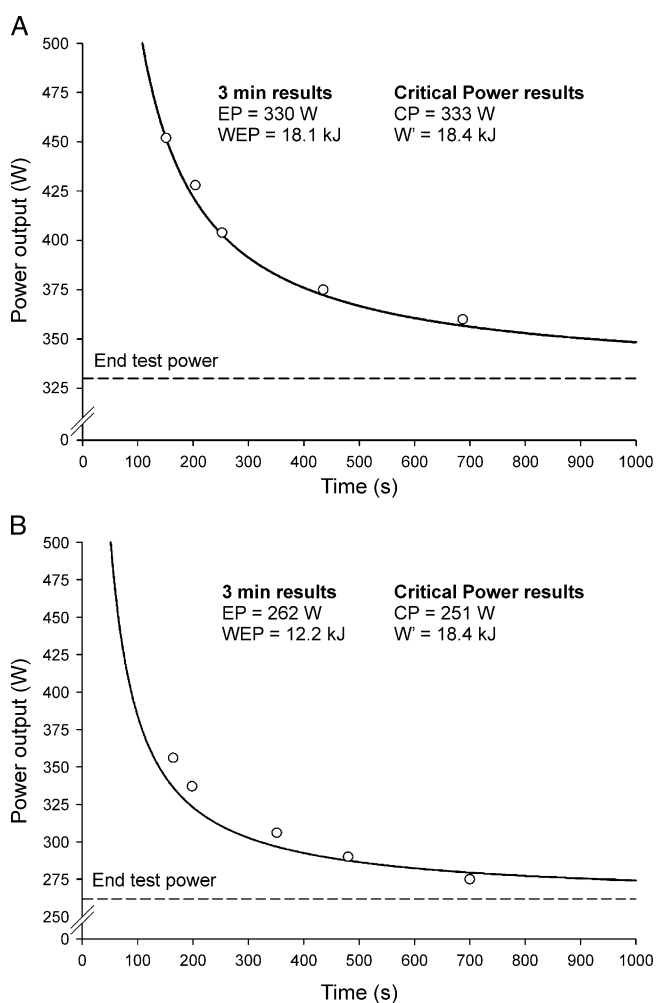


FIGURE 5—The tolerable duration of severe-intensity exercise and the estimation of the power-duration relationship using parameter estimates derived from the 3-min all-out test in two representative subjects, one showing close agreement between the 3-min test and the work-time critical-power model (A), and one in whom both parameters differed appreciably (B). Note that the parameters derived from the 3-min test accurately predict the tolerable duration of constant-work rate exercise in panel A, but they underestimate the tolerable duration of the three highest work rates in panel B.

power–duration relationship lies close to the actual tolerable duration of exercise for this subject, whereas in panel B, for another subject, the tolerable duration is underestimated for the three highest work rates, principally because the WEP underestimates W' in this subject. Within the whole group, the predicted and actual times to exhaustion measured in four of the predicting trials (70 and 80% Δ , and 100 and 105% $\dot{V}O_{2peak}$) were not significantly different ($t_0 = 0.255\text{--}1.789$; $P = 0.107\text{--}0.804$), but because WEP was 1.0 ± 2.6 kJ lower than W' , the actual times to exhaustion tended to be underestimated by the predictions based on the 3-min test parameters (by approximately 11–28 s, on average).

DISCUSSION

The results of the present study support our first hypothesis in showing that the power output in a 3-min all-out cycling test fell to a relatively steady level (EP) that was almost identical to CP. Also consistent with our second hypothesis, WEP was not significantly different from, and was shown to be highly correlated with, the W' parameter. This is the first study to demonstrate that is possible to determine CP and W' using a single bout of all-out exercise.

As a demarcator of heavy and severe exercise intensity domains, the maximal steady-state work rate represents a key determinant of endurance performance. The methods of its establishment (3,16,20,26), however, currently require repeated laboratory visits, which limits their use in exercise testing for both diagnostic and research purposes. To address this issue, we developed the 3-min all-out test, reasoning that this test may provide a valid estimate of the heavy–severe domain boundary (5). Theoretically, the highest power output that can be attained when W' has been depleted is CP (as shown by equation 1). Therefore, during prolonged all-out exercise, the power output should fall towards, and ultimately attain, the CP (5). The present study was designed to test this hypothesis, and the findings herein are consistent with the CP concept (19–21). In short, the present findings show that the power output during a 3-min all-out test declined to reach a steady level within approximately 135 s, and the mean EP in a 3-min all-out test (287 ± 55 W) was the same as the mean CP determined using repeated exhaustive exercise tests (287 ± 56 W; Figs. 2, 3).

The finding of close agreement between the EP and CP develops and extends our previous work (5). Previously, we have shown that the power profile during a 3-min all-out test was highly repeatable and resulted in the attainment of $\dot{V}O_{2peak}$ and a stable power output in the last 30–60 s of the test. Further measurements have shown that in more than 60% of cases, the EP could be used to estimate the maximal steady state (5). Therefore, it is interesting that the results of the present work demonstrate a relatively high degree of success in estimating CP (8 of the 10

subjects returned EP outputs within 5 W of CP), whereas a sizable minority of tests in our previous work did not return a valid estimate of the maximal steady state. These findings could be reconciled by the notion that the estimation of CP may, again in a sizable minority of cases, yield a power output slightly greater than the maximal steady state (4,24,26); if so, it is likely that in our previous work, the EP was equal to the CP.

The second important finding of the present investigation was that although the WEP and W' estimates were not significantly different, WEP was slightly lower than W' in 6 of 10 subjects. The WEP and W' estimates are consistent with estimates in the literature (6,10,22,23), and the WEP should, at least in theory, be equal to W' . Indeed, others, using isokinetic ergometry and a shorter all-out test (90 s), have suggested that this is indeed the case (9). The physiological basis of W' is obscure, but it is generally believed to be representative of an energy store that is used when power output exceeds the CP (6,10,11,21). With regard to the present work, it is possible that the 3-min all-out test duration was not sufficient to deplete W' . However, it has been demonstrated that 90–120 s of all-out exercise is sufficient to accumulate a maximal O_2 deficit (7,9,13,14,30). Moreover, the consistent attainment of an EP equal to the CP after approximately 120–150 s of all-out exercise in the present study (Figs. 1, 5B) and in our previous work (5) suggests that 3 min is an appropriate test duration and that W' was indeed depleted. The small discrepancy between WEP and W' is, therefore, unlikely to have a physiological origin.

The underestimation of W by the WEP was on the order of approximately 1 kJ (Fig. 4)—a relatively small amount of work that, in terms of WEP, can be accumulated within 1–2 s at the initial stages of the 3-min all-out test. Though it cannot be explained by physiological factors, the discrepancy between WEP and W' may be a result of minor errors in power measurement consequent to differences in the acceleration profile of the flywheel during all-out and constant–work rate exercise. That is, flywheel acceleration was present in the first 10 s of the all-out test but was absent in the constant–work rate tests. The use of other types of power measurement that are unaffected by flywheel inertia, such as isokinetic and/or SRM crank measurements (9,29), could reveal the true extent of the WEP parameter in the 3-min all-out test and could, possibly, eradicate the discrepancy in relation to the W' . It is, however, important to stress that flywheel inertia would not have influenced the measurement of the EP output, because the flywheel was not being accelerated beyond the first 10 s of the test.

The findings of the present study show that the peak $\dot{V}O_2$ response to the all-out test was close to (~97%), but significantly lower than, the ramp-determined $\dot{V}O_{2peak}$; this is consistent with the findings of some (14,30) but not others (5,13,29). Although $\dot{V}O_{2peak}$ has been measured using discontinuous and continuous incremental tests for

many years (8,28), it has been suggested that as long as the power output is maintained above the CP, $\dot{V}O_2$ will be driven to its peak value (12). In this regard, all-out exercise may be considered as a possible alternative to incremental or constant-work rate exercise testing for the assessment of $\dot{V}O_{2peak}$ (5,29). The caveat to this assertion is that not all subjects will be able to maintain a high-enough power output for long enough to achieve $\dot{V}O_{2peak}$ (7,14,30) (Table 1). The findings of the present work add weight to the suggestion that although $\dot{V}O_{2peak}$ can be achieved during prolonged all-out exercise (5,13,29), it is by no means certain that it will be achieved. Whether certain individuals (e.g., those with a large anaerobic capacity and/or rapid $\dot{V}O_2$ kinetics) are more likely than others to achieve a true peak $\dot{V}O_2$ during all-out exercise requires further work.

The main practical advantage of the 3-min all-out test over the conventional methods of determining CP is that, using the 3-min all-out method, CP can be established in a single exercise test. Furthermore, establishment of CP from the power profile in an all-out test does not require linear or nonlinear regression analysis—the average power output during the last 30 s is simply recorded. Consequently, errors attributable to extrapolation of a relationship to an asymptote or intercept cannot occur. Moreover, when the

parameter estimates derived from a single all-out test are modeled using the hyperbolic CP equation ($t = WEP/(P - EP)$; Fig. 5), the tolerable exercise time for any power output within the severe-intensity domain can be calculated. The present data show that exercise-time predictions derived using the all-out test and the actual exercise times measured during four of the predicting trials were not significantly different from one another. However, the prediction tended to underestimate the actual exercise time, and this seems to be the direct result of the negative bias in our estimate of the curvature constant of the relationship, that is, the WEP (Figs. 4 and 5B). If the methodological issues concerning the WEP parameter can be solved, then estimation of CP and the accurate prediction of exercise tolerance in the severe domain would be possible in a single test.

In summary, this study has demonstrated that the EP in a 3-min all-out cycling test was essentially identical to CP, whereas WEP in the same test was found to be similar to, but slightly lower than, W' . The findings of the present study indicate that the CP concept can be generalized to all-out exercise. Thus, we suggest that this test offers an advantageous alternative to the conventional protocol of multiple exhaustive exercise tests to determine CP.

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